

STAT 456

ACTUARIAL STATISTICS II

LECTURE 5- Life Insurance: Contingent Payment Models

Dr. Charlotte Chapman Wardy

Department of Statistics and Actuarial Science
University of Ghana

November 29, 2023

This looks at models where a single payment will occur in future based on the occurrence of a defined random event.

Terminology

- b_t benefit function Benefit payable upon the occurrence of an event at time t
- v^t this is the discount function to bring the benefit payable to present value
- z_t this is the present value function it represents $b_t * v^t$
- $E(b_t * v^t)$ known as the **Actuarial Present Value**

Example: Actuarial Present Value

Consider a policy which consists of a payment of \$50,000 contingent upon retirement in 15 years (if the person is still alive). Suppose that the probability of a 45-year old to retire in 15 years is 0.82. That is $_{15}p_{45} = 0.82$. Find the actuarial present value of the contingent payment. Assume, a 6% interest compounded annually.

Insurances Payable at the Moment of Death

- **Continuous Whole Life Insurance:** This is insurance that makes a benefit payment at the time of death of the insured person, whenever that might be.
- It is based on the continuous random variable time $T = T(x)$ (the future life time random variable)
- Let (benefit @ time t) $b_t = 1$ and discount function v^t
Present value random variable:

$$\bar{Z}_x = v^t$$

- The average cost of the insurance is Actuarial Present Value (APV)

$$\bar{A}_x = E(\bar{Z}_x) = \int_0^{\infty} \bar{z}_x f_{\bar{Z}_x}(t) dt = \int_0^{\infty} v^t {}_t p_x \mu(x+t) dt$$

Life Insurance

- The expression under the integral symbol is the benefit of 1 discounted to time $t = 0$ and multiplied by the probability of death in the interval $[t- t+dt]$
- The variability in cost is given by

$$\text{Var}(\bar{Z}_x) = {}^2\bar{A}_x - \bar{A}_x^2$$

where,

$${}^2\bar{A}_x = \int_0^{\infty} v^{2t} {}_t p_x \mu_{x+t} dt$$

Example 1

You are currently 45years old. It is found that your mortality follows De Moivre's law with $\omega = 90$. You purchase a whole life insurance policy that pays a benefit of \$1,000,000 at the moment of death. Calculate the actuarial present value of your death benefits. Assume an annual effective interest rate of 10%.

Example 2

The pdf of a future-lifetime random variable for a life aged x is given by

$$f_T(t) = \mu e^{-\mu t}, \quad t \geq 0.$$

Assume a constant force of interest δ , find $\bar{A}_x$, ${}^2\bar{A}_x$, $\text{Var}(\bar{Z}_x)$

Finite Term Insurance Payable at the moment of death

N- year term life insurance: this is a policy where the payment is made only when the insured person dies within n years of the policy issue. The contingent payment random variable is

$$\bar{Z}_{x:\overline{n}|}^1 = \begin{cases} v^T, & 0 \leq T \leq n \\ 0, & T > n \end{cases}$$

Finite Term Insurance Payable at the moment of death

- where T is the random variable for the time left to live. The cost / APV is denoted below

$$\bar{A}_{x:\overline{n}|}^1$$

- In the symbol above the 1 on x means that 1 unit of money is paid for a life currently aged x if that life dies prior to reaching the age of $x + n$:

Finite Term Insurance Payable at the moment of death

The APV can be expressed as

$$\begin{aligned}\bar{A}_{x:\overline{n}|}^1 &= E(v^T) = \int_0^\infty z f \bar{z}_{x:\overline{n}|}^1(t) dt \\ &= \int_0^n v^t f_T(t) dt = \int_0^n v^t {}_t p_x \mu(x+t) dt\end{aligned}$$

Example

You are given the following information:

- A unit benefit n -year term life insurance policy
- The age at death random variable with exponential parameter μ .
- The constant force of interest δ

Find the Actuarial present value and its variance.

Endowments

- Endowment policies pay the insured only if he /she survives n years. There are 2 main types namely: Pure endowments and Endowment insurance
- **N-year pure endowment:** makes a payment after n years if only the insured person is alive n years after the policy has been issued. For eg. A 30year pure endowment of face value 50,000 Gh issued to life aged x will pay 50,000 in 30 years if x is alive at that time. If x dies before age x+30, x receives nothing. The present value is expressed as

$$\bar{Z}_{x:\overline{n}|}^1 = \begin{cases} 0, & T(x) \leq n \\ v^n, & T(x) > n \end{cases}$$

Endowments N-year pure endowment

$$\begin{aligned}
 E(\bar{Z}_{x:\overline{n}|}^1) &= A_{x:\overline{n}|}^1 \\
 &= 0 * Pr(T \leq n) + V^n * Pr(T > n) \\
 &= v^n {}_n p_x = {}_n E_x
 \end{aligned}$$

There is no bar over A because this is a single discrete payment at time T=n.

$$\begin{aligned}
 Var(\bar{Z}_{x:\overline{n}|}^1) &= {}^2\bar{A}_{x:\overline{n}|}^1 - (\bar{A}_{x:\overline{n}|}^1)^2 \\
 &= v^{2n} {}_n p_x - v^{2n} ({}_n p_x)^2 \\
 &= v^{2n} {}_n p_x {}_n q_x
 \end{aligned}$$

Example

Suppose that the age at death random variable is exponential with constant force of mortality μ . Let $\bar{Z}_{x:\overline{n}|}^1$ be the present value of an n -year pure endowment for a life aged (x) with benefit payment of 1. Assume force of interest δ , find $A_{x:\overline{n}|}^1$, ${}^2A_{x:\overline{n}|}^1$ and $\text{Var}(\bar{Z}_{x:\overline{n}|}^1)$

Endowment Insurance

- N-year endowment insurance makes payment either upon the beneficiary's death or survival to the end of a term n years.
- The contingent payment random variable is given by $\bar{Z}_{x:\overline{n}|}$
- $$\bar{Z}_{x:\overline{n}|} = \begin{cases} v^T, & T(x) \leq n \\ v^n, & T(x) > n \end{cases}$$
- $$\bar{Z}_{x:\overline{n}|} = \bar{Z}_{x:\overline{n}|}^1 + \bar{Z}_{x:\overline{n}|}^{}$$
- $$Var(\bar{Z}_{x:\overline{n}|}) = Var(\bar{Z}_{x:\overline{n}|}^1) + Var(\bar{Z}_{x:\overline{n}|}^{}) + 2Cov(\bar{Z}_{x:\overline{n}|}^1, \bar{Z}_{x:\overline{n}|}^{})$$
- $$\bar{A}_{x:\overline{n}|} = \bar{A}_{x:\overline{n}|}^1 + A_{x:\overline{n}|}^{}$$

Example

The lifetime of a group of people has the following survival function associated with it $s(x) = 1 - \frac{x}{100}$ $0 \leq x \leq 100$. Paul, a member of the group is currently 40 years old and has a 15-year endowment insurance policy, which will pay him \$50,000 upon death. Find the actuarial present value of this policy. Assume an annual force of interest $\delta = 0.05$.

Deferred Life Insurance

An m -year deferred insurance policy pays a benefit to the insured if only the person dies at least m years from the time the policy was issued. The random variable is

$${}_m|\bar{Z}_x = \begin{cases} 0, & T(x) \leq m \\ v^T, & T(x) > m \end{cases}$$

$${}_m|\bar{A}_x = E({}_m|\bar{Z}_x) = E(v^T) = \int_m^{\infty} v^t f_T(t) dt$$

$$= \int_m^{\infty} v^t {}_t p_x \mu_{x+t} dt$$

$$= v^m {}_m p_x \int_m^{\infty} v^{t-m} {}_{t-m} p_{x+m} \mu_{x+m} dt$$

$$= v^m {}_m p_x \int_0^{\infty} v^u {}_u p_{x+m} \mu_{x+m} du + m$$

$$v^m {}_m p_x \bar{A}_{x+m}$$

Example

Let the remaining lifetime at birth random variable X be uniform on $[0, \omega]$. Let ${}_m\bar{Z}_x$ be the contingent payment random variable for a life aged x . Find ${}_m\bar{A}_x$ if the force of interest is δ .

Insurances Payable at the End of the Year of Death

- It is based on $K(x)=k$ which means that death occurred in the interval $(x+k, x+k+1)$.
- $\Pr(K(x)=k) = {}_k p_x q_{x+k}$
- **The discrete Whole Life Insurance** : This is insurance with benefit 1 paid at the end of the year death whenever it may occur.
- The present value random variable for this benefit is $Z_x = v^{K(x)+1}, K(x) \geq 0$
- The Actuarial Present value is

$$A_x = \sum_{k=0}^{\infty} v^{k+1} {}_k p_x q_{x+k}$$

The discrete Whole Life Insurance

- The second moment of Z_x is
- ${}^2A_x = \sum_{k=0}^{\infty} v^{2(k+1)} {}_k p_x q_{x+k}$
- $\text{Var}(Z_x) = {}^2A_x - (A_x)^2$

Discrete Term life Insurance

- The n-year term insurance. This kind has benefit paid to x at the end of year death, after n years
- Present value random variable of benefit 1 is

$$Z_{x:\overline{n}|}^1 = \begin{cases} v^{k+1}, & k = 0, 1, \dots, n-1 \\ 0, & K \geq n \end{cases}$$

$$APV = A_{x:\overline{n}|}^1 = \sum_{k=0}^{n-1} v^{k+1} {}_k p_x q_{x+k}$$

Insurances Payable at the End of the Year of Death

Deferred Life Insurance

Present random variable :

$${}_m|Z_x = \begin{cases} 0, & k = 0, 1, \dots, m-1 \\ v^{K+1}, & K = m, m+1, \dots \end{cases}$$

$$APV = {}_m|A_x = \sum_{k=m}^{\infty} v^{k+1} {}_k p_x q_{x+k}$$

Endowment Life Insurance

Present random variable :

$$Z_{x:\overline{n}|} = \begin{cases} v^{k+1}, & k = 0, 1, \dots, n-1 \\ v^n, & K = n, n+1, \dots \end{cases}$$

Endowment Life Insurance (cont)

$$APV = A_{x:\overline{n}|} = \sum_{k=0}^{n-1} v^{k+1} {}_k p_x q_{x+k} + v^n {}_n p_x$$

Therefore//

$$A_{x:\overline{n}|} = A_{x:\overline{n}|}^1 + A_{x:\overline{n}|}^{}_1$$

Applications of Life Insurance

- **Sufficient Funds for Paying Claims.** An insurance company writes 100 single premium whole life insurances of 10 payable at the moment of death. We assume that the insured have identical independent remaining lifetime random variable. Let us assume that this random variable is exponential with $\mu = 4$ and force of interest $\delta = 0.06$. We would like to know the amount of Fund F that the insurer must have to ensure that the probability of having sufficient funds to pay all the claims is 0.95

Applications of Life Insurance-Sufficient Funds for Paying Claims(cont)

- The contingent random variable is

$$z_i = 10e^{(-0.06t)}, i = 1, 2, \dots, 100$$

Let S be the present value of all future claims. Then

$$S = z_1 + z_2 + \dots + z_{100}$$

- We would like to know the amount of fund F that the insurer must have now to ensure that the probability of having sufficient funds to pay all claims is 0.95

$$Pr(S \leq F) = 0.95 = Pr(z_{0.95} \leq \frac{F - \mu_s}{\delta_s}) \text{ but } z_{0.95} = 1.645$$

Applications of Life Insurance- Sufficient Funds for Paying Claims(cont)

$$F = \mu_s + 1.645\delta_s,$$

$$\mu_s = E(z_i) = 10\bar{A}_x$$

$$= 10\left(\frac{\mu}{\mu + \delta}\right) = 4(\delta^s)^2$$

$$= \text{Var}(z_i) = {}^2\bar{A}_x - (\bar{A}_x)^2$$

$${}^2\bar{A}_x = 100\frac{\mu}{\mu + 2\delta} = 25$$

$$= 25 - 16 = 9$$

$$\delta_s = \sqrt{100\text{Var}(z_i)} = 30$$

$$F = 449.35$$